Miguel Rivera

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EDUCATION

Temple University — College of Engineering | Philadelphia, PA
B.S. in Electrical Engineering — Computer Engineering Concentration | Expected May 2028

Northeast High School | Philadelphia, PA
High School Diploma | June 2024 | GPA: 4.9

TECHNICAL SKILLS

Programming: Python, MATLAB, MakeCode
Engineering: Electrical Circuits, Embedded Systems, CAD, Data Analysis, Mathematical Modeling
Tools: Git, GitHub, VS Code, Microsoft Excel, Linux/Command Line

ENGINEERING PROJECTS

Temperature-Regulated micro:bit Enclosure | Temple University

- Collaborated with two teammates to design and build a self-regulating temperature enclosure using a BBC micro:bit V2, temperature sensors, fans, and external circuitry.
- Developed code to monitor temperature and control the system, integrating hardware and software into a functional prototype.
- Designed and constructed the enclosure using cardboard and Styrofoam and incorporated CAD-designed and 3D-printed components.
- Tested and troubleshot the system to improve temperature regulation and overall functionality.

Wind Energy Analysis & Turbine Modeling | Temple University

- Developed a MATLAB engineering model to analyze wind turbine energy production using Typical Meteorological Year (TMY) weather data.
- Adjusted wind-speed data for turbine hub heights and calculated power and energy production.
- Analyzed monthly energy production and evaluated profitability and breakeven conditions using an economic model.
- Applied MATLAB programming, mathematical modeling, data analysis, and visualization of engineering problems.

RELEVANT COURSEWORK

Introduction to Engineering • Engineering Computation • Calculus • Physics • Chemistry